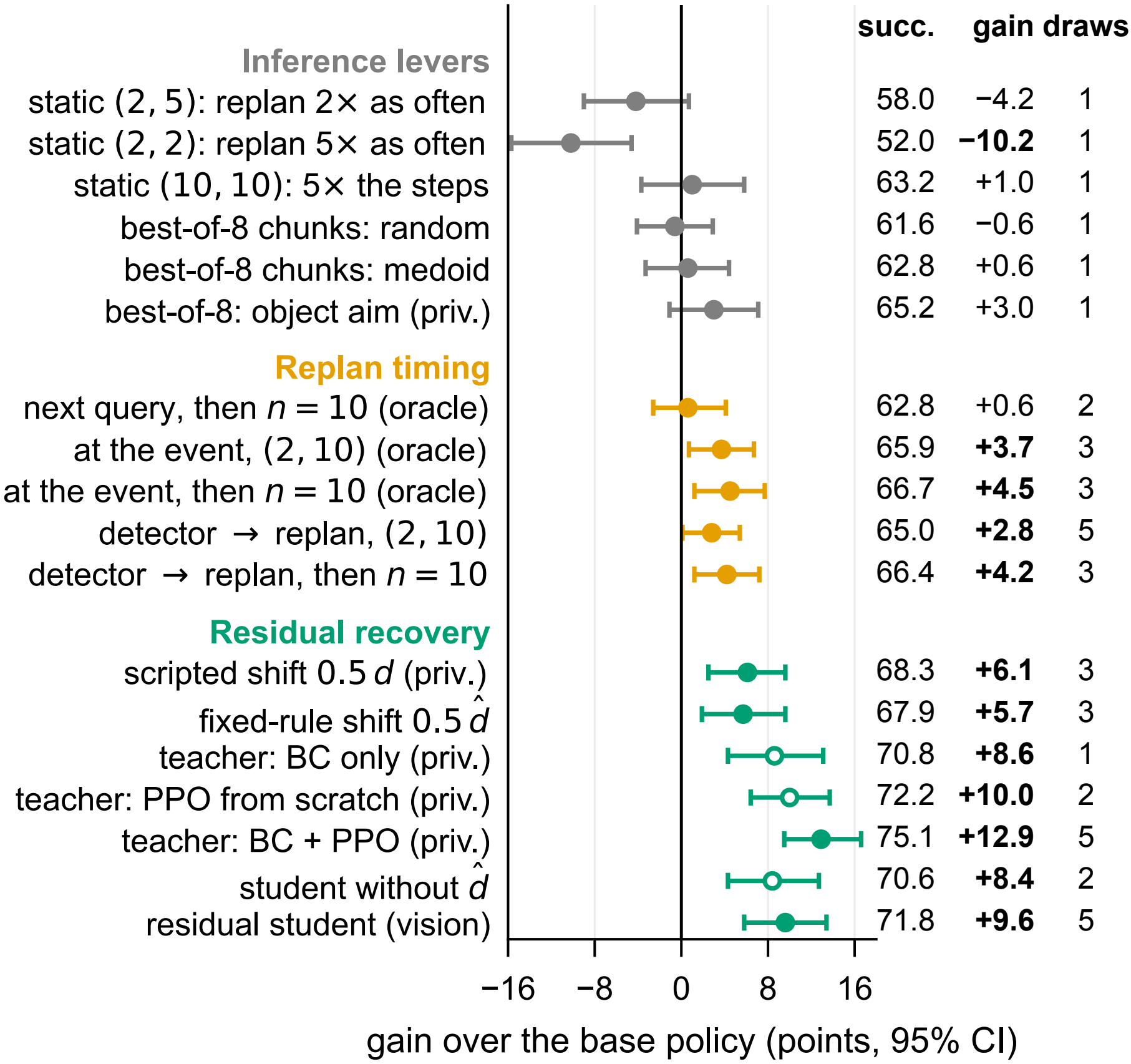
succ.
gain
draws
Inference levers
static (2, 5): replan 2× as often
58.0
−4.2
1
static (2, 2): replan 5× as often
52.0
−10.2
1
static (10, 10): 5× the steps
63.2
+1.0
1
best-of-8 chunks: random
61.6
−0.6
1
best-of-8 chunks: medoid
62.8
+0.6
1
best-of-8: object aim (priv.)
65.2
+3.0
1
Replan timing
next query, then n = 10 (oracle)
62.8
+0.6
2
at the event, (2, 10) (oracle)
65.9
+3.7
3
at the event, then n = 10 (oracle)
66.7
+4.5
3
detector → replan, (2, 10)
65.0
+2.8
5
detector → replan, then n = 10
66.4
+4.2
3
Residual recovery
scripted shift 0.5 d (priv.)
68.3
+6.1
3
fixed-rule shift 0.5 d̂
67.9
+5.7
3
teacher: BC only (priv.)
70.8
+8.6
1
teacher: PPO from scratch (priv.)
72.2
+10.0
2
teacher: BC + PPO (priv.)
75.1
+12.9
5
student without d̂
70.6
+8.4
2
residual student (vision)
71.8
+9.6
5
−16
−8
0
8
16
gain over the base policy (points, 95% CI)